# RAJALAKSHMI ENGINEERING COLLEGE

## DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

### DESIGN THINKING AND INNOVATION

# REVIEW 1 REPORT

### EMPATHIZE AND DEFINE

| | | | |
|---|---|---|---|
| **Academic Year** | 2026- 2027 | **Year / Semester** | III CSD |
| **Section** | A | **Review Date** | 21/ 08 / 2026 |

## TEAM DETAILS

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|---|---|
| **Team Name** | EATALY |
| **Project Title** | In-Campus Food Ordering System (EATALY) |
| **Technology & Domain** | [ ] ML [ ] DL [ ] AI [ ] VR [ ] AR [ ] Animation [ ] Hybrid / Other |
| **Application Area / SDG** | Campus food service (in-campus canteen ordering) |
| **Faculty Coordinator** | Dr. P. Revathy |
| **Team Members** | 1. Roll No: 241701003 Name: AISHWARYA B<br>2. Roll No: 241701024 Name: LATHA A |
| **Team Contact / Email** | |

## REVIEW OBJECTIVE

This review evaluates the team's ability to understand real users and stakeholders, collect authentic evidence, synthesize meaningful insights, and frame a focused human-centred problem. The same process applies to projects using ML, DL, AI, VR, AR, Animation or a hybrid technology; the technology should emerge as a suitable response to a validated need rather than as a predetermined solution.

## PROJECT CONTEXT AND INITIAL SCOPE

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| **Problem Context** | *Where, when and for whom does the problem occur?*<br><br>Occurs on campus, at canteen/cafeteria kiosks (e.g. Hut Cafe and REC Cafe), mainly during peak meal times and short breaks between classes, when students and staff try to place, pay for and collect food orders. |
| **Primary Users / Beneficiaries** | *Identify the main user group(s), not only the client or sponsor.*<br><br>Students who order food during breaks and want to save time by avoiding queues, and college staff/faculty who need quick, convenient food ordering during limited free time. |
| **Key Stakeholders** | *Users, caregivers, teachers, administrators, experts, service providers, decision-makers, etc.*<br><br>Canteen owners/vendors and kitchen staff (Hut Cafe and REC Cafe administrators) who prepare and manage food orders; college administration/management who approve and oversee implementation of the app on campus; app admin(s) who manage canteen listings, monitor orders and control overall platform operations. |
| **Initial Assumption to Validate** | *State the team's starting assumption without presenting a final solution.*<br><br>The team's starting assumption is that the current campus canteen ordering process is inefficient because of long queues, network/payment issues and manual order management, and that students, staff and canteen administrators face difficulty predicting food demand and communicating order status, resulting in delays, overcrowding, food wastage and reduced dining efficiency. |

## USER RESEARCH SUMMARY

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|---|---|
| **Research Methods Used** | [X] Interview [ ] Observation [ ] Survey [ ] Contextual Inquiry [ ] Expert Consultation |
| **Number of Users / Stakeholders** | 5 students and 3 canteen adminstrators |
| **Types of Users / Stakeholders** | Students (primary users)<br>Canteen Administrators (key stakeholders |
| **Research Locations / Platforms** | Campus canteens - Hut Cafe and REC Cafe. |
| **Total Duration of Field Study** | 14 days |
| **Sampling / Selection Rationale** | Students and administrators of the two campus canteens studied (Hut Cafe and REC Cafe) were selected as they are directly involved in, and affected by, the everyday food ordering process. |
| **Important Limitations** | Access, sample diversity, time, privacy, data availability, language, device constraints, etc.<br>• Limited canteen infrastructural facilities: Scaling up the kitchen infrastructure and manpower cannot be achieved rapidly<br>• . Peak hour traffic: The software needs to be able to accommodate thousands of users simultaneously without causing any delay in service provision<br>• .Usability: The software must be user-friendly for students, canteen staff, and administrators having varying degrees of computer literacy skills.<br>• Time and cost constraints: The development of the system has to be kept under certain limits due to the nature of the project. |

## EVIDENCE CHECKLIST

| Evidence Type | Submitted (Yes/No) | File / Annexure Ref. | Faculty Verification / Notes |
|---|---|---|---|
| Research plan and interview / observation guide | No | | |
| Interview notes, transcripts, survey responses or observation records | Yes | Empathy Map slides (Says / Thinks / Does / Feels) for 4 personas | |
| Photographs, screenshots, sketches or field artefacts (where permitted) | Yes | purchase bill photo (Hut Cafe), interview photo (REC Cafe) | |
| Persona / user profile and empathy map | Yes | Empathy Map | |
| Existing-solution / competitor / workflow study | No | | |
| References for secondary research and domain facts | No | | |

## EXISTING SOLUTION / CURRENT WORKFLOW STUDY

| Existing Solution / Practice | What Works Well | Gaps / Unmet Needs |
|---|---|---|
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## RESEARCH INTEGRITY AND INCLUSION CHECK

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|---|---|
| **Privacy and Consent** | *How were identities, photos, recordings and personal data handled responsibly?*<br><br>Photographs of participants were taken during the field study, and participant names were recorded alongside their responses (not anonymized). Participants were informed that their feedback was being collected for an academic design project. |
| **Accessibility and Inclusion** | *Were diverse abilities, languages, devices, connectivity and socio-economic contexts considered?*<br><br>Feedback was collected from students and staff across different departments and years to get varied opinions. We also considered different levels of tech familiarity, especially among canteen staff and admins,. |
| **Bias and Fairness** | *What sampling, data or representation biases may affect the findings?*<br><br>• Sampling may skew toward students who are more vocal about queue frustration, potentially under-representing those who don't mind current wait times.<br>• Canteen staff/vendor perspectives may be underrepresented compared to student perspectives if fewer staff were interviewed.<br>• Convenience sampling (e.g., approaching students already in the canteen queue) may bias findings toward peak-hour experiences and miss off-peak user needs. |
| **Technology Feasibility Signals** | *Only preliminary constraints: data access, hardware, immersion comfort, computational resources, content pipeline, etc.*<br><br>Preliminary constraints identified: the app must work within limited canteen infrastructure (kitchen capacity, staff and equipment); it must function reliably during peak hours with many simultaneous users without crashing or delaying orders; and it must be simple enough for students, canteen staff and admins with varying levels of tech familiarity. |

## INSIGHTS EXTRACTED

| KEY PAIN POINTS / BARRIERS | KEY GAIN POINTS / EXPECTATIONS |
|---|---|
| • Students spend a considerable amount of time waiting in queues to place and collect their food order.<br>• Frequent network and payment issues make the ordering process slow and unreliable.<br>• The absence of a pre-order facility causes overcrowding during peak lunch hours.<br>• Cafe administrators struggle to predict daily food demand, leading to over-preparation or shortages.<br>• Manual order management and limited manpower make it difficult to handle large volumes of customers efficiently.<br>• Lack of real-time communication and inventory tracking results in delays, confusion, and food wastage. | • Students can pre-order meals, reducing waiting time and avoiding long queues.<br>• A faster and more reliable ordering and payment process improves the overall dining experience.<br>• Better scheduling allows students to save time between classes.<br>• Cafe administrators can accurately forecast demand using pre-order data.<br>• Automated order and inventory management reduce workload and improve operational efficiency.<br>• Optimized food preparation minimizes wastage, enhances service quality, and increases customer satisfaction. |

## INSIGHT STATEMENTS

*Write 2–4 evidence-based, non-obvious insights. Each insight should connect an observed behaviour or pattern with an underlying need or motivation.*

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|---|---|
| **1** | Students face long queues along with frequent network and payment issues while ordering, showing a need for a faster, more reliable pre-order and payment system that saves time during short breaks. |
| **2** | The absence of a pre-order facility causes overcrowding during peak lunch hours, showing a need for scheduled, slot-based pickup that spreads out demand across the day. |
| **3** | Cafe administrators struggle to predict daily food demand and rely on manual tracking with limited manpower, showing a need for pre-order data that lets them accurately forecast quantity and reduce over-preparation or shortages. |
| **4** | Poor real-time communication of order status and inventory between students and admins leads to delays, confusion and food wastage, showing a need for automated order and inventory tracking integrated into the app. |

## PROBLEM FRAMING

**POINT OF VIEW (POV) STATEMENT**

[User / stakeholder] needs a way to [need] because [evidence-based insight].

Students and canteen administrators need a way to place, track and manage food orders without long queues or manual guesswork, because network/payment issues, absence of pre-ordering, and manual demand tracking cause up to 20 minutes of waiting time, overcrowding during peak lunch hours, and food wastage from over- or under-preparation, as observed in the empathy maps and interviews of Hut Cafe and REC Cafe administrators and student users.

**HOW MIGHT WE (HMW) STATEMENTS**

1. How might we help students and staff pre-order food and select a convenient pickup time to avoid long kiosk queues?

2. How might we help canteen admins accurately track order volume and predict food quantity needed during peak hours despite limited manpower?

3. How might we enable centralized, real-time management of orders, inventory and demand across multiple campus canteens to reduce delays, overcrowding and food wastage?

## REVIEW 1 SUBMISSION READINESS

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|---|---|
| The project problem is stated from the user / stakeholder perspective, not as a technology feature. | **[X] Yes [ ] No** |
| Primary research evidence is organized and can be verified by the review panel. | **[X] Yes [ ] No** |
| Insights are traceable to research observations, quotations, patterns or artefacts. | **[X] Yes [ ] No** |
| The POV and HMW statements remain solution-neutral and allow multiple concepts. | **[X] Yes [ ] No** |
| Every team member is prepared to explain their individual contribution during viva. | **[X] Yes [ ] No** |

*Team declaration: The submitted field evidence is authentic, ethically collected and accurately represented. Any use of AI tools for transcription, summarisation, translation or visual preparation has been disclosed to the faculty coordinator.*

## GENERALIZED EVALUATION GUIDE – APPLICABLE TO ALL TECHNOLOGY DOMAINS

| Criterion | Max | What the Faculty Evaluates |
|---|---|---|
| **Research Depth & Authenticity** | **5** | Relevant user groups and stakeholders; adequate and diverse primary research; genuine field engagement; context understood; claims not based only on assumptions. |
| **Quality of Evidence Submitted** | **3** | Organized, dated and traceable evidence; suitable notes, transcripts, survey/observation records, photos or artefacts; evidence directly supports findings; privacy respected. |
| **Insight Clarity & Relevance** | **4** | Findings are synthesized into clear pains, gains, needs and opportunity areas; insights are evidence-based, specific, non-obvious and relevant to the selected application context. |
| **POV & HMW Quality** | **3** | POV clearly identifies the user, need and insight. HMW statements are open-ended, actionable, solution-neutral, human-centred and neither too broad nor too narrow. |
| **Understanding of User Research – Viva** | **4** | Student can explain methods, users, evidence, findings, limitations, ethical issues and domain context, and can justify how conclusions were reached. |
| **Clarity in Explaining Insights** | **3** | Student communicates the research-to-insight-to-problem-framing logic clearly, using specific evidence and appropriate terminology. |
| **Contribution & Involvement** | **3** | Student shows verifiable ownership of assigned tasks, participation in field work and synthesis, collaboration, timely documentation and responsible conduct. |

*Scoring note: Award full marks only when the relevant indicators are demonstrated with convincing evidence. Award proportional marks when evidence is partial, weakly connected, generic or unsupported. Technical sophistication alone should not compensate for weak user research or problem framing in Review 1.*

## DOMAIN-NEUTRAL INTERPRETATION NOTES

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|---|---|
| **ML / DL / AI** | Evaluate the human need, decision context, data availability, representation, privacy, bias and explainability signals. Model accuracy or architecture is not the focus of Review 1. |
| **VR / AR** | Evaluate the user setting, device access, interaction needs, safety, motion comfort, spatial constraints, accessibility and the value of immersion or augmentation. |
| **Animation** | Evaluate the intended audience, communication or learning need, narrative comprehension, cultural relevance, accessibility, viewing context and media constraints. |
| **Hybrid Projects** | Apply the same human-centred criteria across the combined technologies. The team must justify why each technology is relevant to the validated need. |
| **All Domains** | Reward evidence-based problem understanding and clear framing. Do not reward a fashionable technology choice when the user need, evidence or opportunity is weak. |

## REVIEW 1 – MARK DISTRIBUTION (25 MARKS PER STUDENT)

### Part A – Team Evaluation (Common Marks) – 15 Marks

| Criteria | Max Marks | Marks Awarded | Brief Faculty Observation |
|---|---|---|---|
| Research Depth & Authenticity | 5 | | |
| Quality of Evidence Submitted | 3 | | |
| Insight Clarity & Relevance | 4 | | |
| POV & HMW Quality | 3 | | |
| **Subtotal – Team Score** | **15** | | |

### Part B – Individual Evaluation – 10 Marks (Per Student)

| Criteria | Max Marks | Student 1 Roll No: _____ | Student 2 Roll No: _____ | Student 3 Roll No: _____ |
|---|---|---|---|---|
| Understanding of User Research (Viva) | 4 | | | |
| Clarity in Explaining Insights | 3 | | | |
| Contribution & Involvement | 3 | | | |
| **Subtotal – Individual Score** | **10** | | | |

## CONSOLIDATED MARKS

| Roll No. | Student Name | Team Score /15 | Individual Score /10 | Total /25 |
|---|---|---|---|---|
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| | | | | |

## COORDINATOR COMMENTS AND ACTION POINTS

| Strengths / Evidence Accepted / Corrections Required / Next Review Expectations | | | |
|---|---|---|---|
| | | | |
| **[ ] Approved** | **[ ] Approved with Revision** | **[ ] Re-review Required** | **Re-review Date: ____ / ____ / 20____** |

**Signature : Faculty Coordinator (Dr. P. Revathy,Professor,CSD)** **Head of the Department**